

Elektrotehnički fakultet Univerziteta u Beogradu

Principi Softverskog Inženjerstva (SI3PSI)

tikimiki

Specifikacija Baze Podatka - Verzija 1.0

digitalci

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1. Uvod

1.1. Namena dokumenta

Baza podataka za projekat tikimiki predstavlja fleksibilan i pouzdan način čuvanja podataka i pristupa istim od strane veb servera radi generisanja veb strana i pružanja funkcionalnosti platforme. U dokumentu se nalazi dijagram sa IE notacijom, šema relacione baze podataka, kao i opis svih tabela u bazi podataka. Ovaj dokument služi kao osnova za razvoj detaljne projektne specifikacije sistema, implementaciju i testiranje.

1.2. Ciljna grupa

Dokument je namenjen vodi (tim lideru) projekta i članovima razvojnog tima. Tim lideru ovaj dokument služi za planiranje razvojnih aktivnosti i specifikaciju imena tabela i polja u bazi, kako bi nezavisne celine, implementirane od strane različitih delova razvojnog tima, na kraju rada bile uspešno integrisane. Razvojnem timu dokument služi kao osnova za dizajn i implementaciju baze podataka.

1.3. Rečnik pojmova i skraćenica

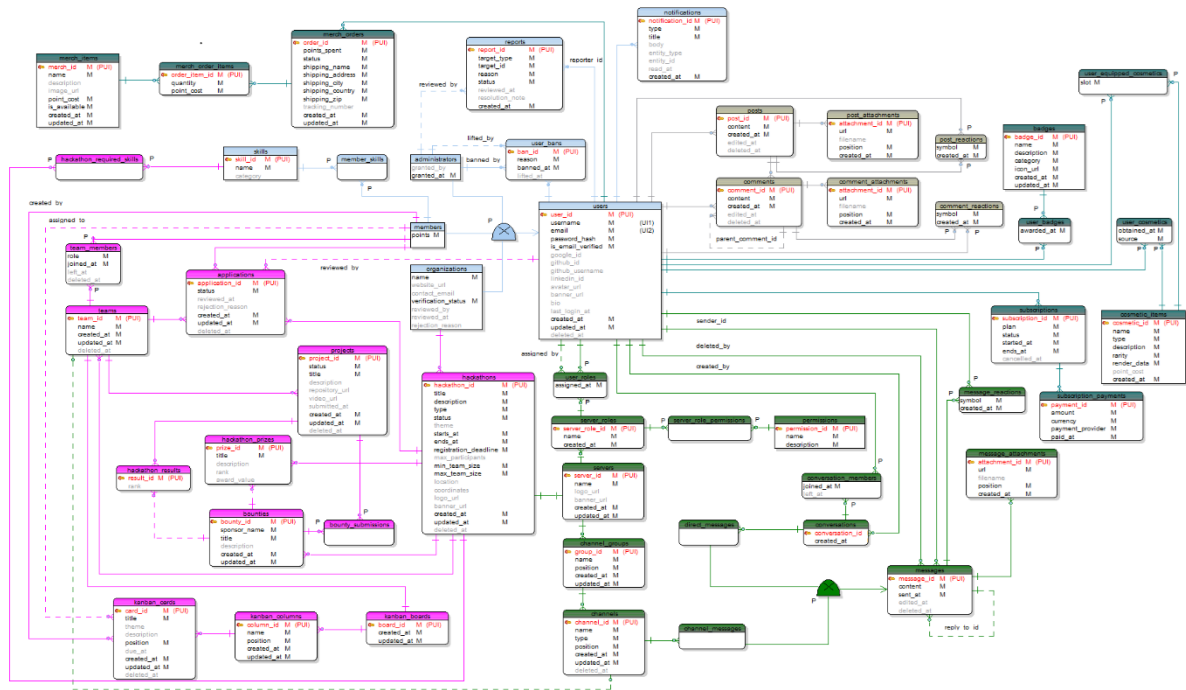
- IE – *Information Engineering*, notacija za modelovanje podataka

1.4. Otvorena pitanja

Broj	Opis	Rešenje

2. Modeli podataka

2.1. Dijagram IE notacije



2.2. Šema relacije baze podataka

users (user_id, username, email, password_hash, is_email_verified, google_id, github_id, github_username, linkedin_id, avatar_url, banner_url, bio, last_login_at, created_at, updated_at, deleted_at)

administrators (user_id, granted_by, granted_at)

members (user_id, points)

organizations (user_id, name, website_url, logo_url, contact_email, verification_status, reviewed_by, reviewed_at, rejection_reason)

user_bans (ban_id, user_id, banned_by, reason, banned_at, lifted_at, lifted_by)

skills (skill_id, name, category)

member_skills (user_id, skill_id)

hackathons (hackathon_id, organization_id, title, description, type, status, theme, starts_at, ends_at, registration_deadline, max_participants, min_team_size, max_team_size, location, coordinates, logo_url, banner_url, created_at, updated_at, deleted_at)

hackathon_required_skills (hackathon_id, skill_id)

bounties (bounty_id, hackathon_id, sponsor_name, title, theme, description, created_at, updated_at)

hackathon_prizes (prize_id, hackathon_id, bounty_id, sponsor_name, title, description, rank, award_value)

teams (team_id, hackathon_id, name, created_at, updated_at, deleted_at)

team_members (team_id, user_id, role, joined_at, left_at, deleted_at)

applications (application_id, user_id, hackathon_id, team_id, status, reviewed_by, reviewed_at, rejection_reason, created_at, updated_at, deleted_at)
projects (project_id, team_id, status, title, description, repository_url, video_url, submitted_at, created_at, updated_at, deleted_at)
hackathon_results (result_id, project_id, bounty_id, rank, prize_id)
bounty_submissions (bounty_id, project_id)
kanban_boards (board_id, team_id, created_at, updated_at)
kanban_columns (column_id, board_id, name, position, created_at, updated_at)
kanban_cards (card_id, column_id, created_by, assigned_to, title, description, position, due_at, created_at, updated_at, deleted_at)
servers (server_id, hackathon_id, name, logo_url, banner_url, created_at, updated_at)
server_roles (server_role_id, server_id, name, created_at)
permissions (permission_id, name, description)
server_role_permissions (server_role_id, permission_id)
user_roles (server_role_id, user_id, assigned_by, assigned_at)
channel_groups (group_id, server_id, name, position, created_at, updated_at)
channels (channel_id, group_id, team_id, type, name, position, created_at, updated_at, deleted_at)
messages (message_id, sender_id, reply_to_id, content, sent_at, edited_at, deleted_at, deleted_by)
message_attachments (attachment_id, message_id, url, filename, position, created_at)
channel_messages (message_id, channel_id)
conversations (conversation_id, created_by, created_at)
conversation_members (conversation_id, user_id, joined_at, left_at)
direct_messages (message_id, conversation_id)
message_reactions (user_id, message_id, symbol, created_at)
posts (post_id, user_id, content, created_at, edited_at, deleted_at)
post_attachments (attachment_id, post_id, url, filename, position, created_at)
post_reactions (user_id, post_id, symbol, created_at)
comments (comment_id, post_id, user_id, parent_comment_id, content, edited_at, created_at, deleted_at)
comment_attachments (attachment_id, comment_id, url, filename, position, created_at)
comment_reactions (user_id, comment_id, symbol, created_at)
badges (badge_id, name, description, category, icon_url, created_at, updated_at)
user_badges (user_id, badge_id, awarded_at)
cosmetic_items (cosmetic_id, type, name, description, rarity, render_data, point_cost, created_at)
user_cosmetics (user_id, cosmetic_id, obtained_at, source)
user_equipped_cosmetics (user_id, slot, cosmetic_id)
merch_items (merch_id, name, description, image_url, point_cost, is_available, created_at, updated_at)

merch_variants (variant_id, merch_id, label, stock)

merch_orders (order_id, user_id, points_spent, status, shipping_name, shipping_address, shipping_city, shipping_country, shipping_zip, tracking_number, created_at, updated_at)

merch_order_items (order_item_id, order_id, merch_id, variant_id, quantity, point_cost)

subscriptions (subscription_id, user_id, plan, status, started_at, ends_at, cancelled_at)

subscription_payments (payment_id, subscription_id, amount_cents, currency, payment_provider, provider_payment_id, paid_at)

reports (report_id, reporter_id, target_type, target_id, reason, status, reviewed_by, reviewed_at, resolution_note, created_at)

notifications (notification_id, user_id, type, title, body, entity_type, entity_id, read_at, created_at)

3. Tabele

3.1. users

Sadrži podatke o korisniku koji se prijavio na platformu.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes		No
username	varchar(32)	No		No
email	varchar(254)	No		No
password_hash	text	No		No
is_email_verified	boolean	No		No
google_id	text	No		Yes
github_id	text	No		Yes
github_username	varchar(39)	No		Yes
linkedin_id	text	No		Yes
avatar_url	text	No		Yes
banner_url	text	No		Yes
bio	text	No		Yes
last_login_at	timestampz	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.2. administrators

Sadrži podatke o administratorima platforme.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
granted_by	uuid	No	administrators	Yes
granted_at	timestampz	No		No

3.3. members

Sadrži podatke o običnom članu platforme.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
points	bigint	No		No

3.4. organizations

Sadrži podatke o organizaciji koja organizuje hakatone na platformi. Atribut verification_status pokazuje status verifikacije organizacije od strane administratora.

org_verification_status = {'pending', 'approved', 'rejected'}

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
name	varchar(100)	No		No
website_url	text	No		Yes
logo_url	text	No		Yes
contact_email	varchar(254)	No		Yes
verification_status	org_verification_status	No		No
reviewed_by	uuid	No	administrators	Yes
reviewed_at	timestampz	No		Yes
rejection_reason	text	No		Yes

3.5. user_bans

Sadrži podatke o korisničkim banovima – sankcijama koje administratori izriču članovima platforme.

Name	Datatype	PK	FK	Nullable
ban_id	uuid	Yes		No
user_id	uuid	No	users	No
banned_by	uuid	No	administrators	No
reason	text	No		No
banned_at	timestampz	No		No
lifted_at	timestampz	No		Yes
lifted_by	uuid	No	administrators	Yes

3.6. skills

Sadrži podatke o veštinama koje članovi mogu posedovati i koje hakatoni mogu zahtevati.

Name	Datatype	PK	FK	Nullable
skill_id	uuid	Yes		No
name	varchar(100)	No		No
category	varchar(50)	No		Yes

3.7. member_skills

Sadrži podatke o veštinama koje poseduju članovi platforme. Predstavlja vezu između tabela members i skills.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	members	No
skill_id	uuid	Yes	skills	No

3.8. hackathons

Sadrži podatke o hackathonima – takmičenjima koja se organizuju na platformi.

hackathon_type = {'physical', 'virtual', 'hybrid'}

hackathon_status = {'upcoming', 'ongoing', 'finished', 'cancelled'}

Name	Datatype	PK	FK	Nullable
hackathon_id	uuid	Yes		No
organization_id	uuid	No	organizations	No
title	varchar(200)	No		No
description	text	No		No
type	hackathon_type	No		No
status	hackathon_status	No		No
theme	varchar(100)	No		Yes
starts_at	timestamptz	No		No
ends_at	timestamptz	No		No
registration_deadline	timestamptz	No		No
max_participants	integer	No		Yes
min_team_size	smallint	No		No
max_team_size	smallint	No		No
location	varchar(200)	No		Yes
coordinates	geography(Point, 4326)	No		Yes
logo_url	text	No		Yes
banner_url	text	No		Yes
created_at	timestamptz	No		No
updated_at	timestamptz	No		No
deleted_at	timestamptz	No		Yes

3.9. hackathon_required_skills

Sadrži podatke o veštinama koje hakaton zahteva od učesnika. Predstavlja vezu između tabela hackathons i skills.

Name	Datatype	PK	FK	Nullable
hackathon_id	uuid	Yes	hackathons	No
skill_id	uuid	Yes	skills	No

3.10. bounties

Sadrži podatke o sponzorskim izazovima (bounty) u okviru hakatona. Svaki izazov pripada jednom hakatonu i ima sponzora.

Name	Datatype	PK	FK	Nullable
bounty_id	uuid	Yes		No
hackathon_id	uuid	No	hackathons	No
sponsor_name	varchar(100)	No		No
title	varchar(200)	No		No
theme	varchar(100)	No		Yes
description	text	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.11. hackathon_prizes

Sadrži podatke o nagradama koje se dodeljuju u okviru hakatona ili pojedinačnog bounty izazova.

Name	Datatype	PK	FK	Nullable
prize_id	uuid	Yes		No
hackathon_id	uuid	No	hackathons	No
bounty_id	uuid	No	bounties	Yes
sponsor_name	varchar(100)	No		Yes
title	varchar(200)	No		No
description	text	No		Yes
rank	smallint	No		Yes
award_value	text	No		Yes

3.12. teams

Sadrži podatke o timovima koji učestvuju na hakatonu. Svaki tim pripada jednom hakatonu.

Name	Datatype	PK	FK	Nullable
team_id	uuid	Yes		No
hackathon_id	uuid	No	hackathons	No
name	varchar(100)	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.13. team_members

Sadrži podatke o članovima jednog tima i njihovim ulogama (leader ili member).

team_role = {'leader', 'member'}

Name	Datatype	PK	FK	Nullable
team_id	uuid	Yes	teams	No
user_id	uuid	Yes	members	No
role	team_role	No		No
joined_at	timestampz	No		No
left_at	timestampz	No		Yes
deleted_at	timestampz	No		Yes

3.14. applications

Sadrži podatke o prijavama članova na hakatone. Atribut status označava trenutni status prijave.

application_status = {'pending', 'approved', 'rejected', 'waitlisted', 'withdrawn'}

Name	Datatype	PK	FK	Nullable
application_id	uuid	Yes		No
user_id	uuid	No	members	No
hackathon_id	uuid	No	hackathons	No
team_id	uuid	No	teams	Yes
status	application_status	No		No
reviewed_by	uuid	No	users	Yes
reviewed_at	timestampz	No		Yes
rejection_reason	text	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.15. projects

Sadrži podatke o projektima koje timovi razvijaju u okviru hakatona. Atribut status opisuje trenutnu fazu projekta.

project_status = {'draft', 'submitted', 'under_review', 'judged'}

Name	Datatype	PK	FK	Nullable
project_id	uuid	Yes		No
team_id	uuid	No	teams	No
status	project_status	No		No
title	varchar(200)	No		No
description	text	No		Yes
repository_url	text	No		Yes
video_url	text	No		Yes
submitted_at	timestampz	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.16. hackathon_results

Sadrži podatke o rezultatima projekata na hakatonu i pripadajućim nagradama. Ako je bounty_id NULL, rezultat se odnosi na generalno rangiranje.

Name	Datatype	PK	FK	Nullable
result_id	uuid	Yes		No
project_id	uuid	No	projects	No
bounty_id	uuid	No	bounties	Yes
rank	integer	No		Yes
prize_id	uuid	No	hackathon_prizes	Yes

3.17. bounty_submissions

Sadrži veze između projekata i bounty izazova na koje su projekti predati.

Name	Datatype	PK	FK	Nullable
bounty_id	uuid	Yes	bounties	No
project_id	uuid	Yes	projects	No

3.18. kanban_boards

Sadrži podatke o Kanban tabli koju koristi tim za vođenje projekta. Svaki tim ima najviše jednu tablu.

Name	Datatype	PK	FK	Nullable
board_id	uuid	Yes		No
team_id	uuid	No	teams	No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.19. kanban_columns

Sadrži podatke o kolonama na Kanban tabli.

Name	Datatype	PK	FK	Nullable
column_id	uuid	Yes		No
board_id	uuid	No	kanban_boards	No
name	varchar(100)	No		No
position	real	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.20. kanban_cards

Sadrži podatke o pojedinačnim karticama (zadacima) na Kanban tabli.

Name	Datatype	PK	FK	Nullable
card_id	uuid	Yes		No
column_id	uuid	No	kanban_columns	No
created_by	uuid	No	members	No
assigned_to	uuid	No	members	Yes
title	varchar(200)	No		No
description	text	No		Yes
position	real	No		No
due_at	timestampz	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.21. servers

Sadrži podatke o serveru za komunikaciju koji se vezuje za hakaton (slično Discord serveru).

Name	Datatype	PK	FK	Nullable
server_id	uuid	Yes		No
hackathon_id	uuid	No	hackathons	No
name	varchar(200)	No		No
logo_url	text	No		Yes
banner_url	text	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.22. server_roles

Sadrži podatke o ulogama koje postoje na nekom serveru za komunikaciju.

Name	Datatype	PK	FK	Nullable
server_role_id	uuid	Yes		No
server_id	uuid	No	servers	No
name	varchar(100)	No		No
created_at	timestampz	No		No

3.23. permissions

Sadrži podatke o dozvolama koje se mogu dodeliti ulogama na serveru.

Name	Datatype	PK	FK	Nullable
permission_id	uuid	Yes		No
name	varchar(100)	No		No
description	text	No		No

3.24. server_role_permissions

Sadrži vezu između uloga i dozvola – koje dozvole su dodeljene kojoj ulozi.

Name	Datatype	PK	FK	Nullable
server_role_id	uuid	Yes	server_roles	No
permission_id	uuid	Yes	permissions	No

3.25. user_roles

Sadrži podatke o tome koja uloga je dodeljena kom korisniku na serveru.

Name	Datatype	PK	FK	Nullable
server_role_id	uuid	Yes	server_roles	No
user_id	uuid	Yes	members	No
assigned_by	uuid	No	users	Yes
assigned_at	timestampz	No		No

3.26. channel_groups

Sadrži podatke o grupama kanala (kategorijama) u okviru servera. Atribut position definiše redosled.

Name	Datatype	PK	FK	Nullable
group_id	uuid	Yes		No
server_id	uuid	No	servers	No
name	varchar(100)	No		No
position	real	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.27. channels

Sadrži podatke o kanalima na serveru. Kanal može biti opšti, za obaveštenja, timski ili privatni.

channel_type = {'general', 'announcements', 'team', 'private'}

Name	Datatype	PK	FK	Nullable
channel_id	uuid	Yes		No
group_id	uuid	No	channel_grou ps	No
team_id	uuid	No	teams	Yes
type	channel_type	No		No
name	varchar(100)	No		No
position	real	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No
deleted_at	timestampz	No		Yes

3.28. messages

Sadrži podatke o porukama koje korisnici razmenjuju (kako u kanalima, tako i u direktnim razgovorima). Atribut `reply_to_id` omogućava odgovor na poruku.

Name	Datatype	PK	FK	Nullable
message_id	uuid	Yes		No
sender_id	uuid	No	users	No
reply_to_id	uuid	No	messages	Yes
content	text	No		No
sent_at	timestampz	No		No
edited_at	timestampz	No		Yes
deleted_at	timestampz	No		Yes
deleted_by	uuid	No	users	Yes

3.29. message_attachments

Sadrži podatke o prilogima (slike, dokumenti) koji su priloženi uz poruku.

Name	Datatype	PK	FK	Nullable
attachment_id	uuid	Yes		No
message_id	uuid	No	messages	No
url	text	No		No
filename	varchar(255)	No		Yes
position	smallint	No		No
created_at	timestampz	No		No

3.30. channel_messages

Veza koja označava da je poruka poslata u određeni kanal servera.

Name	Datatype	PK	FK	Nullable
message_id	uuid	Yes	messages	No
channel_id	uuid	No	channels	No

3.31. conversations

Sadrži podatke o privatnim konverzacijama (direktnim porukama) između korisnika.

Name	Datatype	PK	FK	Nullable
conversation_id	uuid	Yes		No
created_by	uuid	No	users	No
created_at	timestampz	No		No

3.32. conversation_members

Sadrži podatke o učesnicima konverzacije (jedan-na-jedan ili grupne direktne poruke).

Name	Datatype	PK	FK	Nullable
conversation_id	uuid	Yes	conversations	No
user_id	uuid	Yes	users	No
joined_at	timestampz	No		No
left_at	timestampz	No		Yes

3.33. direct_messages

Veza koja označava da je poruka deo privatne konverzacije, a ne kanala servera.

Name	Datatype	PK	FK	Nullable
message_id	uuid	Yes	messages	No
conversation_id	uuid	No	conversations	No

3.34. message_reactions

Sadrži reakcije korisnika (emodži) na poruke. Atribut symbol predstavlja emodži.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
message_id	uuid	Yes	messages	No
symbol	text	Yes		No
created_at	timestampz	No		No

3.35. posts

Sadrži podatke o objavama koje korisnici postavljaju na društvenom feedu platforme.

Name	Datatype	PK	FK	Nullable
post_id	uuid	Yes		No
user_id	uuid	No	users	No
content	text	No		No
created_at	timestampz	No		No
edited_at	timestampz	No		Yes
deleted_at	timestampz	No		Yes

3.36. post_attachments

Sadrži podatke o priložima (slike, dokumenti) priloženim uz objavu.

Name	Datatype	PK	FK	Nullable
attachment_id	uuid	Yes		No
post_id	uuid	No	posts	No
url	text	No		No
filename	varchar(255)	No		Yes
position	smallint	No		No
created_at	timestampz	No		No

3.37. post_reactions

Sadrži reakcije korisnika (emodži) na objave.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	users	No
post_id	uuid	Yes	posts	No
symbol	text	Yes		No
created_at	timestampz	No		No

3.38. comments

Sadrži podatke o komentarima ostavljenim na objavama. Atribut `parent_comment_id` omogućava ugnježdene odgovore.

Name	Datatype	PK	FK	Nullable
<code>comment_id</code>	uuid	Yes		No
<code>post_id</code>	uuid	No	posts	No
<code>user_id</code>	uuid	No	users	No
<code>parent_comment_id</code>	uuid	No	comments	Yes
<code>content</code>	text	No		No
<code>edited_at</code>	timestampz	No		Yes
<code>created_at</code>	timestampz	No		No
<code>deleted_at</code>	timestampz	No		Yes

3.39. comment_attachments

Sadrži podatke o prilozima priloženim uz komentar.

Name	Datatype	PK	FK	Nullable
<code>attachment_id</code>	uuid	Yes		No
<code>comment_id</code>	uuid	No	comments	No
<code>url</code>	text	No		No
<code>filename</code>	varchar(255)	No		Yes
<code>position</code>	smallint	No		No
<code>created_at</code>	timestampz	No		No

3.40. comment_reactions

Sadrži reakcije korisnika (emodži) na komentare.

Name	Datatype	PK	FK	Nullable
<code>user_id</code>	uuid	Yes	users	No
<code>comment_id</code>	uuid	Yes	comments	No
<code>symbol</code>	text	Yes		No
<code>created_at</code>	timestampz	No		No

3.41. badges

Sadrži podatke o bedževima koji se dodeljuju članovima.

badge_category = {'participation', 'achievement', 'social', 'special'}

Name	Datatype	PK	FK	Nullable
badge_id	uuid	Yes		No
name	varchar(100)	No		No
description	text	No		No
category	badge_category	No		No
icon_url	text	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.42. user_badges

Sadrži podatke o bedževima koje je član osvojio i datumu dodele.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	members	No
badge_id	uuid	Yes	badges	No
awarded_at	timestampz	No		No

3.43. cosmetic_items

Sadrži podatke o kozmetičkim predmetima (efekti za korisničko ime, dekoracije avatara, efekti banera).

cosmetic_type = {'username_effect', 'avatar_decoration', 'banner_effect'}

cosmetic_rarity = {'common', 'rare', 'epic', 'legendary'}

Name	Datatype	PK	FK	Nullable
cosmetic_id	uuid	Yes		No
type	cosmetic_type	No		No
name	varchar(100)	No		No
description	text	No		Yes
rarity	cosmetic_rarity	No		No
render_data	jsonb	No		No
point_cost	integer	No		Yes
created_at	timestampz	No		No

3.44. user_cosmetics

Sadrži podatke o kozmetičkim predmetima koje je član stekao. Atribut source označava način sticanja.

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	members	No
cosmetic_id	uuid	Yes	cosmetic_items	No
obtained_at	timestampz	No		No
source	varchar(50)	No		No

3.45. user_equipped_cosmetics

Sadrži podatke o trenutno postavljenim kozmetičkim predmetima člana po slotu (tipu).

Name	Datatype	PK	FK	Nullable
user_id	uuid	Yes	members	No
slot	cosmetic_type	Yes		No
cosmetic_id	uuid	No	cosmetic_items	No

3.46. merch_items

Sadrži podatke o merchandise predmetima koji se mogu kupiti za poene. Atribut is_available pokazuje da li je predmet trenutno dostupan.

Name	Datatype	PK	FK	Nullable
merch_id	uuid	Yes		No
name	varchar(200)	No		No
description	text	No		Yes
image_url	text	No		Yes
point_cost	integer	No		No
is_available	boolean	No		No
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.47. merch_variants

Sadrži podatke o varijantama merchandise predmeta (npr. veličina, boja) i njihovim stanjima na zalihama.

Name	Datatype	PK	FK	Nullable
variant_id	uuid	Yes		No
merch_id	uuid	No	merch_items	No
label	varchar(100)	No		No
stock	integer	No		No

3.48. merch_orders

Sadrži podatke o porudžbinama merchandise predmeta i adresi za isporuku. Atribut points_spent predstavlja broj utrošenih poena.

merch_order_status = {'pending', 'processing', 'shipped', 'delivered', 'cancelled'}

Name	Datatype	PK	FK	Nullable
order_id	uuid	Yes		No
user_id	uuid	No	members	No
points_spent	integer	No		No
status	merch_order_status	No		No
shipping_name	varchar(200)	No		No
shipping_address	text	No		No
shipping_city	varchar(100)	No		No
shipping_country	char(2)	No		No
shipping_zip	varchar(20)	No		No
tracking_number	varchar(100)	No		Yes
created_at	timestampz	No		No
updated_at	timestampz	No		No

3.49. merch_order_items

Sadrži podatke o pojedinačnim stavkama u okviru jedne merchandise porudžbine. Atribut `point_cost` je cena u poenima u trenutku porudžbine.

Name	Datatype	PK	FK	Nullable
order_item_id	uuid	Yes		No
order_id	uuid	No	merch_orders	No
merch_id	uuid	No	merch_items	No
variant_id	uuid	No	merch_variants	Yes
quantity	smallint	No		No
point_cost	integer	No		No

3.50. subscriptions

Sadrži podatke o premium pretplatama korisnika.

`subscription_plan = {'premium'}`

`subscription_status = {'active', 'cancelled', 'expired'}`

Name	Datatype	PK	FK	Nullable
subscription_id	uuid	Yes		No
user_id	uuid	No	members	No
plan	subscription_plan	No		No
status	subscription_status	No		No
started_at	timestampz	No		No
ends_at	timestampz	No		No
cancelled_at	timestampz	No		Yes

3.51. subscription_payments

Sadrži podatke o pojedinačnim uplatama vezanim za pretplate. Atribut `amount_cents` je iznos u centima izabrane valute.

Name	Datatype	PK	FK	Nullable
payment_id	uuid	Yes		No
subscription_id	uuid	No	subscriptions	No
amount_cents	integer	No		No
currency	char(3)	No		No
payment_provider	varchar(50)	No		No
provider_payment_id	text	No		No
paid_at	timestampz	No		No

3.52. reports

Sadrži podatke o prijavama neprikladnog sadržaja ili neregularnosti. Atribut `target_type` i `target_id` čine polimorfnu referencu na prijavljenu jedinicu.

`report_target_type = {'user', 'post', 'comment', 'message', 'hackathon'}`

`report_status = {'pending', 'reviewed', 'resolved', 'dismissed'}`

Name	Datatype	PK	FK	Nullable
report_id	uuid	Yes		No
reporter_id	uuid	No	users	No
target_type	report_target_type	No		No
target_id	uuid	No		No
reason	text	No		No
status	report_status	No		No
reviewed_by	uuid	No	administrators	Yes
reviewed_at	timestampz	No		Yes
resolution_note	text	No		Yes
created_at	timestampz	No		No

3.53. notifications

Sadrži podatke o obaveštenjima koja se prikazuju korisnicima. Atributi `entity_type` i `entity_id` čine polimorfnu referencu na povezan entitet.

```
notification_type = {'application_approved', 'application_rejected',
                    'application_waitlisted', 'badge_awarded', 'hackathon_result_posted',
                    'hackathon_starting_soon', 'organization_verified',
                    'organization_rejected', 'new_direct_message', 'position_assigned',
                    'bounty_result_posted', 'merch_order_shipped'}
```

```
entity_type = {'user', 'hackathon', 'application', 'team', 'project', 'post',
               'comment', 'badge', 'message', 'bounty'}
```

Name	Datatype	PK	FK	Nullable
notification_id	uuid	Yes		No
user_id	uuid	No	users	No
type	notification_type	No		No
title	varchar(100)	No		No
body	text	No		Yes
entity_type	entity_type	No		Yes
entity_id	uuid	No		Yes
read_at	timestampz	No		Yes
created_at	timestampz	No		No

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Alfa	17.05.2026.	Alfa verzija	Andrej Čolić
1.0	21.05.2026.	Kompletna verzija	Andrej Čolić